

Radioactive Substances (Use and Regulation) Act, 2020

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Chapter-1

Preliminary

1. Short Title and Commencement: (1) This Act may be called the "Radioactive Material (Use and Regulation) Act, 2077".

(2) This Act shall commence on the thirty-first day from the date of its certification. 2. Definitions: Unless the subject or context otherwise requires, in this Act,-

(a) "Practice" means any human activity that increases exposure or potential exposure to additional radiation from radioactive materials and other radiation sources, or alters the network of exposure sources or pathways or pathways.

(b) "Ionizing Radiation" means radiation capable of producing ion pairs in biological material.

(c) "License" means a license issued pursuant to Section 11 to perform a specific task in relation to any practice, activity, or facility.

(d) "Licensee" means an entity that has obtained a license pursuant to Section 11.

(e) "Agency" means the International Atomic Energy Agency.

(f) "Activity" means actions related to the production, use, export, import, and transport of ionizing radiation sources, the siting, construction, commissioning, operation, and decommissioning or closure of facilities, radioactive waste management, and site restoration.

(g) "Transport" means the carriage of radioactive material from the place of production or storage to another place, and that word also includes the activities of preparing packages for such transport, loading onto a means of transport, storage during transit, and unloading at the final destination.

(h) "Prescribed" or "as prescribed" means prescribed or as prescribed in the rules made under this Act.

(i) "Export" means the act of exporting radioactive material and related equipment or technology out of Nepal.

(j) "Regulatory Body" means the regulatory body pursuant to Section 4.

(k) "Directive Committee" means the directive committee pursuant to Section 6.

(l) "Inspector" means a person appointed from the Regulatory Body pursuant to subsection (2) of Section 22.

(m) "Import" means the act of importing radioactive material, related equipment or technology into Nepal.

(n) "Ministry" means the Ministry of the Government of Nepal dealing with science and technology subjects.

(o) "Radioactive Material" means material capable of producing radiation such as alpha particles or beta particles or gamma rays.

- (p) "Radioactive Waste" means radioactive material discharged from any practice, activity, or facility that cannot be used in its current state and emits ionizing radiation in an amount exceeding the prescribed level, or any object or article contaminated together with it.
- (q) "Radioactive Source" means radioactive material and equipment capable of emitting or spreading ionizing radiation.
- (r) "Emergency" means a risk or danger arising due to exposure to ionizing radiation occurring in any practice or activity.
- (s) "Disposal" means the safe disposal or displacement of used fuel or radioactive waste without re-use.
- (t) "Safety" means measures adopted to protect human beings and the environment from radiation risks.
- (u) "Source" means any object, material, or equipment capable of emitting or spreading ionizing radiation.
- (v) "Facility" means a place for the production, processing, use, storage, or disposal of radioactive material, and that word also includes the infrastructure requiring the adoption of protective measures against ionizing radiation.
- (w) "Entity" means an entity established or registered under prevailing law.
5. Functions, Duties, and Powers of the Regulatory Body: In addition to the functions, duties, and powers mentioned elsewhere in this Act, the functions, duties, and powers of the Regulatory Body shall be as follows:-
- (a) Formulate necessary plans and programs for the regulation of practices, activities, or facilities related to radioactive sources,
- (b) Prepare necessary standards for the protection of human beings, animals, property, and the environment from the possible adverse or harmful effects of ionizing radiation,
- (c) Determine the conditions, obligations, and responsibilities to be fulfilled by a licensee for conducting any activity or practice,
- (d) Prescribe the qualifications of technical personnel working in a licensed entity,
- (e) Prepare a national record of licensees and radiation sources and regularly update such record,
- (f) Subject to this Act, issue, suspend, or revoke licenses for operating radioactive sources and related activities, practices, or facilities,
- (g) Monitor and inspect whether this Act, the rules, directives, and standards made under this Act, and the conditions and circumstances specified in the license are complied with,
- (h) Inspect, monitor, and investigate the activities, practices, or facilities operated by a licensee,
- (i) Take action under this Act if, during inspection, monitoring, and investigation pursuant to clauses (g) and (h), it is found that this Act and the conditions specified in the license are not complied with,
- (j) Exchange information, experiences, and cooperation with international organizations, or similar institutions or agencies of other countries,
- (k) Conduct awareness programs for the general public and stakeholders regarding ionizing radiation and

accidents that may occur from its use,

(l) Adopt necessary security measures to prevent misuse or unauthorized use of radioactive sources and facilities,

(m) Cooperate, collaborate, and coordinate with concerned government agencies and stakeholders in the formulation and implementation of plans relating to the preparedness for the management of disasters related to radioactive sources,

(n) Conduct or cause to conduct studies and research related to the safety and protection of ionizing radiation,

(o) Make necessary coordination with the Agency. 6.

Directive Committee: (1) There shall be a Directive Committee as follows to provide necessary directions to the Regulatory Body and the Nuclear Research Centre in relation to radioactive materials:- (a) Minister, dealing with Science and Technology - Chairperson (b) Chief of Army Staff, Nepali Army - Member (c) Secretary, Ministry of Industry, Commerce and Supplies - Member (d) Secretary, Ministry of Agriculture and Livestock Development - Member (e) Secretary, Ministry of Home Affairs - Member (f) Secretary, Ministry - Member (g) Secretary, Ministry of Defence - Member (h) Secretary, Ministry of Health and Population - Member (i) Vice-Chancellor, Nepal Academy of Science and Technology - Member (j) Inspector General, Nepal Police - Member (k) Inspector General, Armed Police Force, Nepal - Member (l) Three experts, including at least one woman, nominated by the Ministry - Member (m) Joint Secretary, Ministry - Member-Secretary (2) The term of office of the members pursuant to clause (l) of subsection (1) shall generally be two years. (3) Notwithstanding anything contained in subsection (2), if any nominated member fails to fulfill their official responsibilities, they may be removed from office at any time. Provided that, before such removal, they shall be given an opportunity to defend themselves. (4) The Secretariat of the Directive Committee shall be located at the Ministry. 7. Functions, Duties and Powers of the Directive Committee: The functions, duties and powers of the Directive Committee shall be as follows:- (a) Carry out overall evaluation and review of the use of radioactive materials and the study and research work of nuclear materials, (b) Facilitate the resolution of any obstruction, hindrance, or ambiguity arising in the implementation of the law relating to radioactive materials and their use or practice, (c) Give directions to the Regulatory Body regarding activities related to the control of pollution that may arise from radioactive radiation, (d) Provide necessary directions to the Regulatory Body and the Nuclear Research Centre regarding the control or mitigation of any adverse effect that may be seen to affect the overall peace and security of the country, the life and property of the general public, and the environment from the use or practice of radioactive materials and nuclear research work, (e) Provide suggestions to the Government of Nepal regarding the effective implementation of security measures as per any treaty to which Nepal is a party or an agreement made with the Agency, (f) Perform other functions as prescribed. 8.

Meeting of the Directive Committee: (1) The meeting of the Directive Committee shall be held at least twice a year. (2) The meeting of the Directive Committee shall be held at the date, time, and place fixed by the Chairperson. (3) The Member-Secretary of the Directive Committee shall give notice to all members, including the agenda of subjects to be discussed in the meeting, at least twenty-four hours before the meeting. (4) If more than fifty percent of the members are present in the Directive Committee, the quorum shall be deemed to be constituted. (5) The opinion of the majority shall prevail in the meeting of the Directive Committee, and in case of a tie, the Chairperson shall cast the deciding vote. (6) The Member-Secretary shall cause the decision of the meeting of the Directive Committee to be certified by the Chairperson and kept on

record. (7) Other procedures relating to the meeting of the Directive Committee shall be as determined by the Directive Committee itself. Chapter-3 Provisions relating to License 9. License Required: (1) No entity shall perform any work related to radioactive sources and related practices, activities, or facilities without obtaining a license from the Regulatory Body. (2) An entity having the prescribed technical and professional capacity, desiring to conduct practices, activities, or facilities related to radioactive sources for peaceful purposes, shall submit an application to the Regulatory Body in the prescribed format. (3) Notwithstanding anything contained elsewhere in this section, any individual or entity that, at the time of commencement of this Act, has obtained permission from the concerned body to operate practices, activities, or facilities related to radioactive sources and is operating them, shall obtain a license under this Act within one year from the date of commencement of this Act. 10. Inquiry into Application: (1) The Regulatory Body shall conduct necessary inquiries into the application submitted pursuant to Section 9. (2) While conducting an inquiry pursuant to subsection (1), if it is deemed necessary to conduct a site inspection and obtain additional information or details, the Regulatory Body may request additional information or details from the applicant. 11.

Issuance of License: (1) If, upon inquiry pursuant to Section 10, the Regulatory Body finds such practice, activity, or facility suitable for operation, a license shall be issued in the prescribed format after collecting the prescribed deposit and fee in a manner capable of bearing the possible loss, damage, or risk. (2) While issuing a license pursuant to subsection (1), the Regulatory Body may also specify additional conditions to be fulfilled by the licensee, considering the nature of such practice or activity. (3) If, upon inquiry pursuant to Section 10 into the application received under Section 9, it is not found feasible to issue a license, the Regulatory Body shall give written information thereof to the applicant within seven days from the date of decision. 12. License Not Transferable: A licensee shall not sell, distribute, or transfer the license obtained under this Act. 13. Surrender of License: (1) A licensee may submit an application to the Regulatory Body in the prescribed format for the surrender of a license taken to operate any practice, activity, or facility under this Act. (2) The Regulatory Body shall approve the surrender of a license under subsection (1) if the surrender does not appear to adversely affect the protection of the life and property of the general public, animals, or the environment, and on the condition that if it is confirmed in the future that the life and property of the general public or the environment are affected by the work and cause of that licensee, the licensee shall bear the liability thereof. 14. Amendment of License Conditions: (1) If it is deemed necessary to amend any condition specified in a license issued under this Act for the protection of health, safety, and the environment, the Regulatory Body may amend such condition by giving written notice to the licensee. (2) If, due to the amendment of a condition specified in the license under subsection (1), the licensee needs to make any additional arrangements or preparations, the Regulatory Body shall give a reasonable time to the concerned licensee for this purpose. 15. License Period and Renewal: (1) The period of a license shall be as prescribed. (2) The license shall be renewed from the Regulatory Body, along with the prescribed fee, before the expiry of the period under subsection (1). (3) Other provisions relating to license renewal shall be as prescribed. 16.

Suspension of License: (1) The Regulatory Body may suspend a license for a specified period in the following circumstances:- (a) If the entity obtaining the license fails to fulfill the conditions, comply with the standards and obligations to be observed, or fails to comply with the order given pursuant to subsection (6) of Section 22, or (b) If, due to the activities of the entity obtaining the license, an adverse effect is caused to the environment, creating a risk to the safety of the life and property of the general public or the protection of the environment. (2) Within the suspension period specified by the Regulatory Body, the licensee may submit an application to the Regulatory Body for the revocation of the suspension, stating that the condition under clause (a) or (b) of subsection (1) does not exist. (3) Upon receipt of an application from the licensee under subsection (2), the Regulatory Body may revoke the suspension if, upon inquiry, the condition of subsection

(1) is found not to exist. 17. Revocation of License: (1) The Regulatory Body shall revoke a license in the following circumstances:- (a) If any act is done contrary to this Act, or the regulations or directives made under this Act, (b) If necessary corrections are not made within the suspension period, the conditions specified in the license are violated, or such conditions are not fulfilled, (c) If the license is not renewed within the prescribed period, (d) If the activity or practice is not started or cannot be started within the prescribed period of the license, or (e) If the license is surrendered. (2) Except in the circumstances of clauses (c), (d), and (e) of subsection (1), the Regulatory Body shall give the concerned licensee a reasonable time to submit their defense before revoking the license. 18. May Lodge Complaint: (1) An entity dissatisfied with the decision made by the Regulatory Body pursuant to subsection (3) of Section 11, subsection (1) of Section 16, or subsection (1) of Section 17 may lodge a complaint with the Ministry within thirty-five days from the date of such decision. (2) The decision made by the Ministry on the complaint lodged under subsection (1) shall be final. (3) Lodging a complaint under subsection (1) shall not exempt the licensee from the obligation to be fulfilled under this Act. 19.

Liability of Licensee: (1) In addition to the matters mentioned elsewhere in this Act, the liability of the licensee shall be as follows:- (a) Operate the practice, activity, or facility according to the license in a safe, reliable, and trustworthy manner in accordance with approved standards, (b) Comply with this Act or rules, directives, standards, conditions, and circumstances of the license made under this Act, (c) Provide information, documents, details, or knowledge requested by the Regulatory Body within the prescribed period, (d) Provide necessary assistance to the Regulatory Body during monitoring and inspection, (e) Comply with the conditions and dose limits prescribed regarding radiation protection or safety, (f) Ensure that the amount of radiation released into the environment remains at a minimal level, (g) Make safe arrangements for the decommissioning or destruction of facilities and the disposal of waste, (h) If the practice, activity, or facility under the license needs to be closed, obtain approval from the Regulatory Body before closing such practice, activity, or facility, (i) Fulfill other obligations prescribed by the Regulatory Body from time to time. (2) In addition to the liabilities under subsection (1), the licensee shall be liable for any loss or damage caused due to an incident arising from ionizing radiation. (3) If loss or damage is caused due to stolen, lost, discarded, or abandoned radioactive material due to failure to make reliable security arrangements or due to negligent storage, the licensee shall be liable for such loss or damage. (4) Notwithstanding anything contained in subsections (1) and (2), the Licensee shall not be liable for loss or damage arising from an accident caused by war, external attack, or armed rebellion. (5) If loss or damage is caused due to an incident arising from ionizing radiation during the transport of radioactive material, the liability shall be as per a separate written agreement between the licensee and the transporter, and in the absence of an agreement, the licensee shall be liable for such loss or damage. 20. Prior Approval Required: If the licensee needs to make any change in the activity or practice or address mentioned in the license, prior approval of the Regulatory Body shall be obtained. 21.

Chapter-4

Provisions relating to Regulation

22. Monitoring and Inspection: (1) The Regulatory Body shall regularly monitor and inspect or cause to be monitored and inspected whether this Act, the regulations and directives made under this Act, and the conditions and circumstances of the license are complied with.

(2) To carry out monitoring and inspection under subsection (1), the Regulatory Body may, as required, designate or appoint Government of Nepal employees or subject matter experts having prescribed

qualifications and training as Inspectors.

(3) An Inspector designated or appointed under subsection (2) may inspect the following matters of a licensed individual or entity:-

(a) Enter the place or building where the practice, activity, or facility related to radioactive sources is operated at any time to obtain information about radiation safety and protection,

(b) Obtain information about preparedness for precautions if there is a possibility of an accident at the place or building where the practice, activity, or facility related to radioactive sources is operated,

(c) Investigate and probe any unforeseen incident or accident involving radiation sources,

(d) Make inquiries to any concerned person as necessary.

(4) The Regulatory Body may appoint an Inspector to the licensed entity for monitoring if it appears that conditions and standards prescribed in this Act or the rules made under this Act have not been complied with, or an unexpected incident or unauthorized activity has occurred or may occur, or if information is received from any reliable source.

(5) It shall be the duty of the concerned licensee to assist the Inspector appointed for monitoring and inspection under this Act.

(6) If, during inspection under this section, it appears that a practice, activity, or facility violates this Act, regulations, directives, standards made under this Act, license conditions, or other conditions determined by the Regulatory Body, or that human health or animals may be harmed, or property or the environment may be damaged due to it, the Inspector may order such licensee to comply with the conditions within a specified period, to immediately stop or suspend such practice, activity, or facility, or to store in a safe and reliable manner the radioactive material produced from the practice, activity, or facility that has been ordered to be stopped.

(7) If an order is given by the Inspector under subsection (6), information thereof shall be immediately given to the Regulatory Body.

(8) If an order is given by the Inspector under subsection (6), an inspection report shall be submitted to the Regulatory Body as prescribed.

(9) If, from the report received under subsection (8), it appears that serious damage to public health or the environment may occur, the Regulatory Body may take possession of such nuclear material or radioactive material and facility.

23. Regulation over Radioactive Source: (1) The Regulatory Body shall prescribe necessary standards for the purpose of regulating radioactive sources, materials, and related instruments and equipment.

(2) The Regulatory Body shall classify such sources as prescribed based on the loss or damage that may be caused to the general public and the environment due to the lack of safe management or protection of the source.

(3) If, while using any radioactive source, the licensee sees a risk that radiation damage may be caused to human health or animals, or loss or damage may be caused to property or the environment, the licensee shall inform the Regulatory Body thereof. 24. Record of Radioactive Source: (1) The Regulatory Body shall maintain a record of radioactive sources.

(2) Other provisions relating to maintaining a record of radioactive sources shall be as prescribed. 25. Safety and Protection of Radioactive Source: (1) The responsibility for the safety, protection, and reliable use of radioactive sources shall lie with the licensee.

(2) Other provisions relating to the safety and protection of radioactive sources shall be as prescribed. 26. Export or Import of Radioactive Source: The Government of Nepal shall, on the recommendation of the Regulatory Body, prescribe the conditions and procedures to be fulfilled to obtain a license for the purpose of exporting or importing radioactive sources into Nepal or through the territory of Nepal. 27. Management of Orphan Source: If any orphan source is found, the Regulatory Body shall immediately take control of such source and make proper management thereof. Chapter-5

Provisions relating to Radiation Safety and Protection

28. Radiation Safety: (1) The Regulatory Body shall prescribe necessary standards and measures to protect human health or animals or the environment from loss or damage that may occur due to exposure to ionizing radiation.

(2) The Regulatory Body shall determine the maximum limit of radiation dose for individuals when operating any practice, activity, or facility under this Act.

(3) While determining the limit under subsection (2), the Regulatory Body may also take into account the standards of the Agency and other recognized international institutions.

(4) The Regulatory Body shall identify sources or practices that do not require regulation on the following basis:-

(a) The level of risk for an individual is found to be extremely low compared to that which requires regulation,

(b) The total radiological impact is extremely low, making regulatory control unnecessary. 29.

Radiation Protection: (1) The Regulatory Body shall determine conditions relating to the protection of radioactive sources to be fulfilled before granting a license to operate any practice, activity, or facility. (2) In addition to the conditions determined under subsection (1) regarding the protection of radioactive sources for operating practice, activity, or facility relating to agriculture, animal husbandry, industry, and medical service, the Regulatory Body shall also prescribe the following conditions:- (a) Required qualifications and training of the radioactive source user, (b) Measures to be adopted for the protection of individuals using radioactive sources, (c) Measures to be adopted for the protection of patients, including justification of the practice and optimization of exposure, (d) Standards relating to the design and operation of radioactive sources, (e) Measures to be adopted for the safety and protection of radioactive sources, (f) Provisions relating to not being allowed to conduct health examinations of any patient, provide medical services to a patient, or diagnose or treat a disease using radiation sources except upon a physician's recommendation. (3) Other provisions relating to conditions for the protection of radioactive sources shall be as prescribed. Chapter-6

Provisions relating to Decommissioning or Decontamination of Facility 30. Decommissioning or Decontamination of Facility: The Regulatory Body shall prescribe necessary conditions and standards for the decommissioning or decontamination of any facility and radioactive material. 31. Responsibility for Decommissioning or Decontamination of Facility: (1) The responsibility for safety and protection while closing, decommissioning, or decontaminating any facility or radioactive material shall lie with the licensee. (2) The licensee shall obtain approval from the Regulatory Body before permanently closing, decommissioning, or decontaminating any facility. (3) While granting approval under subsection (2), the Regulatory Body shall

specify conditions including the safe closure, decommissioning, or decontamination of the facility and the management of waste generated therefrom. (4) The expenses incurred for closing, decommissioning, or decontaminating a facility under subsection (2) and managing the waste generated therefrom shall be borne by the licensee. Chapter-7 Provisions relating to Disaster Preparedness and Disaster Management Plan 32.

Disaster Preparedness and Disaster Management Plan: (1) The Government of Nepal shall prepare a disaster preparedness and disaster management plan at the national level for the preparedness and risk management of radiation-related disasters. (2) Subject to subsection (1), a licensee operating any activity, practice, or facility, or keeping or using a radiation source, shall prepare a disaster preparedness and disaster management plan for radiation-related disasters and get it approved by the Regulatory Body. (3) Matters to be included while preparing the disaster preparedness and disaster management plan under subsection (2) shall be as prescribed by the Regulatory Body. (4) The disaster preparedness and disaster management plan approved under subsection (2) shall be reviewed and updated from time to time by the Regulatory Body. (5) The disaster preparedness and disaster management plan approved under subsection (2) shall be implemented. (6) It shall be the duty of all bodies and officials of the provincial government and local level to assist in implementing the disaster preparedness and disaster management plan under subsection (1). (7) The Government of Nepal may establish a fund with the participation of the licensee for compensation for loss or damage arising from practices or activities, and the management and operation of that fund shall be as prescribed. 33. Regarding Transboundary Disaster: (1) If a radiation-related disaster occurs and there is a possibility that the effect of radioactive pollution may spread beyond the state of Nepal, the Government of Nepal shall inform the Agency and the concerned bodies of the country affected or likely to be affected by such radiation. (2) If a radiation-related disaster occurs or may occur from outside the state of Nepal, the Government of Nepal shall immediately take diplomatic initiative regarding disaster management with the Agency and the concerned nation. (3) The Regulatory Body shall act as a contact point for providing information about and necessary assistance regarding such disaster, in accordance with the provisions of the relevant treaty relating to providing information and rendering assistance in the event of a radiation-related disaster. Chapter-8 Provisions relating to Transport of Radioactive Material 34.

Conditions and Standards to be Made: (1) Subject to this Act, the Regulatory Body shall formulate and enforce conditions and standards to be complied with for the safe transport of radioactive material. (2) The conditions and standards under subsection (1) shall also specify measures to be adopted for the physical protection of such radioactive material, based on the risk or harm that may occur according to the type, quantity, and level of activity of the radioactive material. (3) While determining conditions and standards under subsection (1), the technical standards determined by the Agency and developed and followed internationally regarding the transport of radioactive material shall also be taken into account. 35. Transport of Radioactive Material: (1) While transporting radioactive material, the licensee shall do so safely in a manner that does not adversely affect public health, the environment, and national security. (2) The licensee shall transport radioactive material according to the means, mode, and technical conditions of transport determined by the Regulatory Body. (3) The licensee shall inform the workers and employees involved in the transport of radioactive material about the safe use, safety, and physical protection of radioactive material, as well as the risks from such material and possible measures to avoid them. Chapter-9 Provisions relating to Radioactive Waste Management 36. Import Not Allowed: Notwithstanding anything contained elsewhere in this Act, radioactive waste shall not be imported within the borders of Nepal. 37. Responsibility for Safety of Radioactive Waste: (1) The responsibility for the safety management of radioactive waste shall lie with the licensee. (2) The licensee shall make necessary arrangements so that as little radioactive waste as possible is discharged. (3) The licensee shall manage or dispose of radioactive waste as prescribed at a place or

facility approved by the Regulatory Body. 38.

Export of Radioactive Waste: (1) Any radioactive waste produced in Nepal may be exported from Nepal with permission from the Regulatory Body. (2) An individual or entity desiring to export radioactive waste shall submit an application to the Regulatory Body for permission to export, stating the following details:- (a) That the importing country has the administrative and technical capacity for radioactive waste management and has a reliable and safe regulatory framework consistent with the standards implemented by the Agency, (b) That the country importing the radioactive waste has been informed of such import of radioactive waste and that country has provided consent for it, (c) That assurance has been given that transport will be carried out through transit countries in a manner consistent with relevant international obligations. (3) If the process under subsection (2) appears to have been completed, the Regulatory Body may grant a license to the applicant to export radioactive waste. 39. Safety and Protection: To protect human health, animals, and the environment from adverse effects of radioactive waste, the Regulatory Body shall formulate and implement necessary standards. Chapter-10 Provisions relating to Safeguards 40. Implementation of Safeguards: (1) The Regulatory Body shall implement safeguards in accordance with any treaty to which Nepal is a party or any safeguards agreement made with the Agency. (2) While implementing safeguards under subsection (1), the following acts may be performed in accordance with the agreement:- (a) Inspect the implementation of Nepal's obligations, (b) Facilitate inspectors appointed by the Agency within Nepal, (c) Coordinate with the Agency as per the agreement and provide necessary information. 41. Prohibited Acts: The production and manufacture of radiation-related weapons and explosive devices within Nepal, or otherwise obtaining such weapons or devices, accepting or receiving any assistance in the manufacture of such weapons or devices, conducting any research and development related to the production of such weapons or devices, or performing any act that would support such activity shall not be permitted. 42. Inspection May Be Conducted: An inspector appointed by the Regulatory Body or the Agency in coordination with the Regulatory Body shall have the right to enter any prescribed facility in accordance with the safeguards agreement and conduct inspection and inquiry. 43. Records and Control: Provisions relating to the measurement or calibration of prescribed radioactive materials, evaluation of the accuracy of measurement or calibration, review of differences in measurement or calibration, inventory keeping, evaluation of unmeasured items, recording and reporting, and reporting to the Agency shall be as prescribed. 44.

Export or Import of Radioactive Material: (1) No person shall export or import, or cause to export or import, radioactive material, information, instruments, equipment, and technology related to such material without obtaining a license in accordance with prevailing law. (2) The conditions, procedures, standards, and other provisions for issuing a license to export or import radioactive material under subsection (1) shall be in accordance with prevailing law. Chapter-11 Provisions relating to Safety and Physical Protection of Radioactive Material 45. Physical Protection: (1) The Regulatory Body may classify radioactive materials according to security sensitivity and standards for the physical protection of radioactive materials and facilities. (2) The Regulatory Body shall determine and implement conditions and standards for the safety and physical protection of radioactive material on the following matters:- (a) Protection mechanisms and procedures required for different types of materials, (b) Relating to control and accountability, (c) Recording and control of radioactive material, (d) Relating to inspection and monitoring, (e) Other matters as prescribed. (3) The duty of guaranteeing the physical protection of radioactive materials and facilities shall lie with the licensee. (4) If radioactive material or a facility is stolen, lost, or damaged through theft or otherwise, or there is a possibility thereof, the licensee shall perform the following actions:- (a) Immediately inform the Regulatory Body about such incident and the related circumstances, (b) Submit a written report with detailed particulars to the Regulatory Body within three days of providing information under clause (a), (c) Provide other

information as requested by the Regulatory Body within the prescribed time. (5) The Regulatory Body shall immediately provide the information received under subsection (4) to the Ministry. 46.

Chapter-12

Provisions relating to Offense and Punishment

47. Deemed to have Committed Offense: If any person performs the following acts, they shall be deemed to have committed an offense under this Act:-

(a) Conducting or causing to conduct activities or practices including the production, collection, storage, accumulation, export, import, transport, carriage, operation, disposal, and management of equipment generating ionizing radiation without a license under this Act or contrary to the conditions specified in the license or this Act,

(b) Conducting or causing to conduct activities or practices including the production, collection, storage, accumulation, export, import, transport, carriage, disposal, or dissemination of radioactive material or related equipment, developing or operating any facility, exporting, importing, or managing radioactive waste without a license under this Act or contrary to the conditions specified in the license or this Act,

(c) Obtaining, taking possession of, using, transferring, altering, disposing of, or disseminating radioactive material, or making, using, or taking possession of any instrument or equipment with the intention of killing someone, causing physical injury, or damaging property or the environment, or in a manner that would cause death, injury, or damage,

(d) Using any facility, tampering with it, causing loss or damage to it, obstructing its operation, or performing any act targeted against the facility, with the intention of killing someone, causing physical injury, or damaging property or the environment, or knowingly that death, injury, or damage may result,

(e) Stealing, robbing, or looting radioactive material,

(f) Using coercion or fraudulent dealings to obtain radioactive material,

(g) Performing acts under clauses (c) and (e) with the objective of forcing any individual or entity to do or not do something,

(h) Demanding radioactive material or related equipment, instruments, or facilities from anyone by giving fear, threat, or intimidation, or by using force or any coercion,

(i) Spreading or causing to spread false information regarding radioactive material in a manner that may negatively affect the use, physical protection, and safety of radioactive material or related equipment,

(j) Disclosing information that must be kept confidential under this Act. 48.

Punishment: (1) If any person commits an offense under this Act, they shall be liable to the following punishment:- (a) For an offense under clause (a) of Section 47, confiscation of the equipment generating ionizing radiation and a fine of one hundred thousand rupees to three hundred thousand rupees or imprisonment for one year to three years or both, (b) For an offense under clause (b) of Section 47, confiscation of the radioactive material and a fine of seven hundred thousand rupees to one million two hundred thousand rupees and imprisonment for five years to ten years, (c) If an offense under clauses (c) and (d) of Section 47 results in someone's death, the offender shall be liable to life imprisonment; if someone is injured or physically harmed, a fine of five hundred thousand rupees to one million rupees and

imprisonment for five years to ten years, (d) For an offense under clauses (e), (f), (g), (h), (i), and (j) of Section 47, confiscation of the equipment, instruments, other materials, and items used in such offense, and a fine of three hundred thousand rupees to one million rupees and imprisonment for three years to five years. (2) If any person orders or abets the commission of an offense mentioned in Section 47, the offender shall be liable to the same punishment as the principal offender, and an accomplice who assists in committing the offense shall be liable to half the punishment of such principal offender. (3) If any physical property or the environment is damaged due to an incident related to ionizing radiation, the actual loss caused thereby shall be identified, double the amount of the loss shall be compensated by the licensee, a fine equal to the amount of such loss shall be imposed along with imprisonment for five years to ten years, and the equipment, other materials, and items used in such act shall be confiscated. (4) If any licensee commits an offense punishable under this